Méridien Editorial:
Toward Continuity by Max Bill

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Continuity, a colossal 4.5-meter-high granite sculpture shaped as an infinity loop, located in central Frankfurt, Germany. It was created by Swiss architect and polymath Max Bill (1908–1994), a prominent figure in modernism and concrete art.

Imagine a 200-ton block of granite, extracted from a quarry in Sardinia in the 1980s, carved into a work of art and shipped across the North Atlantic, then transported through Europe via canals and rivers, before finally being installed as a monumental sculpture in the form of a smooth, continuous infinity loop in the heart of Frankfurt's financial district, Germany. Such an ambitious undertaking, combining production and logistics, had not been seen since the transport of colossal sculptures carved from monoliths in ancient Egypt. So what was the idea behind it, and what made it possible forty years ago?

1. Introduction to the sculpture (description, 3D overview).
2. Max Bill's short introduction (achievements, influence).
3. The journey from Sardinien to Frankfurt (request and support of Deutsche Bank, chronics)
4. The making process (model, challenges, Dominique Stroobant's role and the team work).

5. The mathematics of the sculpture (model, topology, calculations).
+ Walk around (Gaussian splatting interactive scene)

1. A Sculpture Continuity

“In Continuity, I attempted to realize an idea that had long been present in my mind: the representation of an infinite space in infinite motion. Subsequently, I engaged with the problems realized in it, which are extraordinarily fascinating. That sculpture consists of a band of uniform width throughout, whose edges are therefore parallel. It thus appears to be a two-sided surface, separated from the surrounding space by parallel lines and forming a new space. The phenomenon of this “body,” however, is that this new space-forming surface has only a single side, and that the seemingly parallel two lines are in fact only one single line—that is, a line running parallel to itself.”

In Continuity, I intended to achieve something similar: a surface, a boundary. Purely theoretically, from a twin band—one that is twisted one and a half times around itself and fused with another that is twisted only half a turn—one could obtain a form similar to the “endless loop,” but with only one surface and one boundary. For reasons unknown to me, this did not come about. Thus Continuity, despite a theoretically different expected result, consists of two parallel, bounded surfaces, composed of infinitely many straight lines.

This explanation may seem somewhat dry and primitive, yet it is the only thing I can say about it myself. I can, however, add that for me such a sculpture represents something like **a “knot in space,” a point of concentration of various latent directions, and that through its**

existence space is ordered, newly formed, and given aim and direction.

Facts:
Year: 1986
Place: Frankfurt-am-Main, Germany
Hight: 4,5 m
Weight: 80 t
Production: 3 years
Material: Granit (Sardenia)
Author: Max Bill
Chief of production: Dominique Stroobant
Budget: 2,3M Euro* (*equivalent of 1986 DM to 2026 Euro)

2. About Max Bill

Max Bill (1908–1994) was a great Swiss polymath: an architect, industrial and graphic designer, sculptor, painter, and also a prolific writer, scholar, and teacher. attended the Bauhaus where he was taught by Josef Albers, László Moholy-Nagy, Paul Klee, Wassily Kandinsky and Oskar Schlemmer. Bill remained closely associated with the Bauhaus school and was a key figure in developing and propagating its principles, as a guest lecturer at the Kunstgewerbeschule Zürich and as a co-founder and first rector of the Ulm School of Design.

(texts)

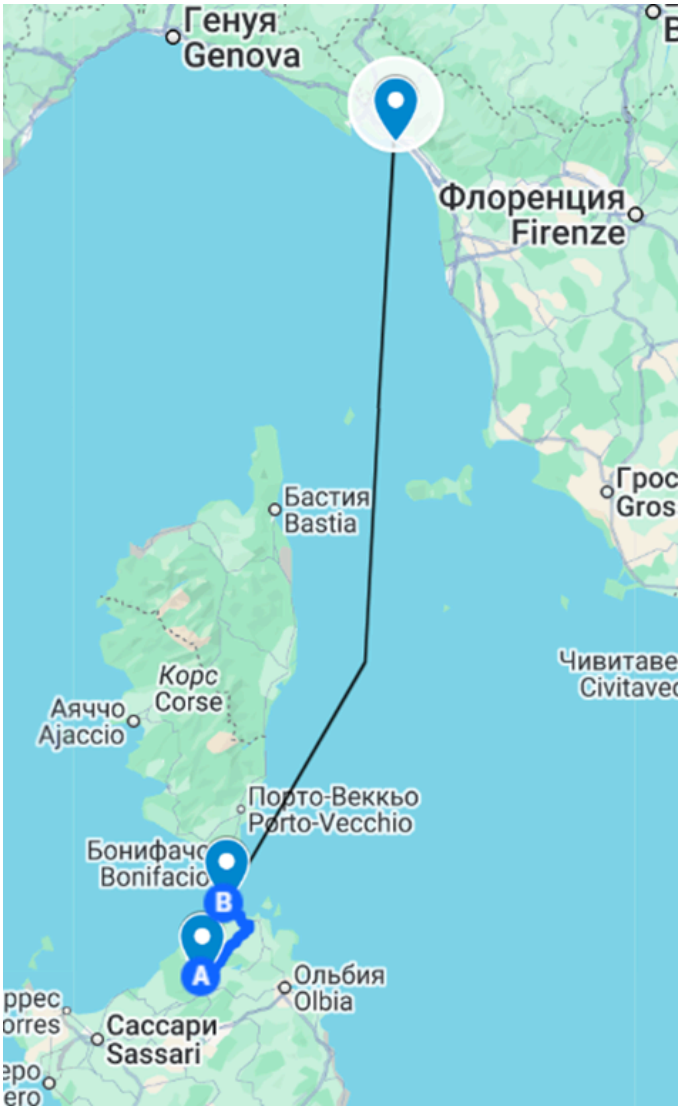
3. A journey from Sardinian quarry to Frankfurt's financial district

— The route and the granite peace;

— Transportation from the quarry to the workshop;
— The journey across Europe.

Facts:
Distance: 2800km by water; 1200km roads
Time: 6 months
Countries: 5

Granite Trip
Quarry Scarraciana → Porto di Santa Teresa Gallura → Porto Marina di Carrara



25.04.1983: The "Esagono" work collective is formed → 13.09.1983: Max Bill and Rodolfo Costantino inspect the Scarraciana quarry → 20.10.1983: A granite block is blasted and extracted (500 tons). → October–November 1983: the block is reduced to 300 tons, then to 180 tons, formed into a polyhedron, and loaded onto a Fagioli truck with a 52-meter platform → A temporary road is built to the highway → 8–9.02.1984: the road is blocked for two days, a huge vehicle carries the block across Sardinia to the port of San Teresa di Gallura → 10-day wait due to weather conditions → 20.02.1984: the ship arrives at the port of Marina di Carrara

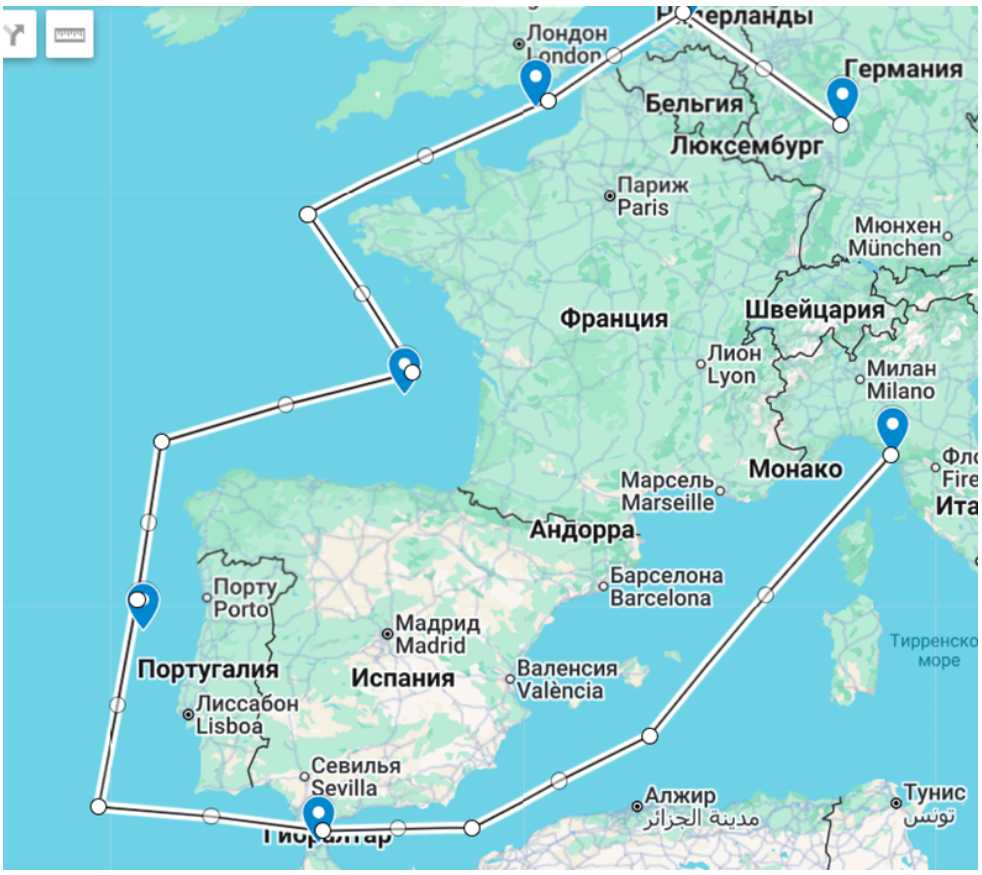
Sculpting

- Work began: February 1984
- A location near the port was chosen for convenient stone transportation and waste storage (approximately 120 tons).
- A temporary metal building and a separate pavilion for the model were constructed.
- Measurements were taken using a grid of special "measuring points" transferred from a four-fold magnified model using giant dividers
- The carving was carried out using: drilling machines and specially made diamond chains (regular ones are used only for marble).
- The sculpture was carved from the inside using complex and dangerous methods.
- The work had to be carried out in shifts: only two people could work at a time.
- Gradually, long internal channels necessary for the shape of the object were cut out.
- By January 1985, the main through-cuts were completed. On March 22, 1986, the sculpture was turned over and placed on three support points, and work continued.

- On June 30, 1986, the sculpture was completed

Sculpture Trip

Port of Carrara → Gibraltar → Portugal → Bay of Biscay → English Channel → Rotterdam → (via the Rhine and Main rivers) → Frankfurt am Main



Marina di Carrara, Italy

4–8 August 1986 – The sculpture was mounted into a steel transport frame for shipment to the port.
12–13 August 1986 – The sculpture was loaded onto the cargo ship *Kupari*.

Sea route

From Marina di Carrara, the ship set out on the following route:
→ through the **Strait of Gibraltar**,
→ along the coast of **Portugal**,
→ across the **Bay of Biscay**,
→ through the **English Channel**,
→ to **Rotterdam, the Netherlands**.

The sea journey lasted from **16 August to 2 September 1986**.

Rotterdam, the Netherlands

– In Rotterdam, the sculpture was unloaded from the seagoing vessel in bad weather.
– It was then transferred onto a **river barge** for transport via inland waterways.

River route

From Rotterdam, the sculpture travelled along the **Rhine**, then the **Main**, to **Frankfurt am Main, Germany**.

Frankfurt am Main, Germany

– At Frankfurt's **Eastern Harbour (Osthafen)**, a crane unloaded the sculpture from the barge and placed it onto a transport platform.
– On the night of **7 September 1986**, it was transported through the city to **Taunusanlage 12**.
– The following morning, a crane installed the **80–90-ton sculpture** on the prepared foundations beside the **Deutsche Bank towers**.

4. From granite block to sculpture. A team challenge.

In 1981, Max Bill met the young and energetic sculptor and photographer Dominique Stroobant. This encounter grew into a lifelong friendship. Their greatest challenge would become the project Continuity, the development of which took three years. Stroobant was responsible for the entire production process: from planning the transportation route of a massive block of granite to the creation of the full-scale, four-meter sculpture together with his team of four sculptors.

Dominique Stroobant is a Belgian artist and mathematician known for his work at the intersection of geometry, perception, and conceptual art. Closely associated with the legacy of Max Bill, Stroobant contributed to projects that extend Bill's mathematically grounded approach to form. His practice explores spatial structures, continuity, and transformation—often translating abstract mathematical principles into precise visual or sculptural expressions. Stroobant is particularly recognized for his involvement in the realization and interpretation of complex geometric artworks, where calculation and craftsmanship are inseparable.

5. Mathematic behind the artwork

Max Bill was consistently inspired by mathematics and logic in art. His approach, which he called "verifiable logic," guided him from early experimentation toward a precise method of making—and explaining—his work: "You can always understand my work, even if you don't like it." From the beginning of his career, Bill was fascinated by the Möbius strip. From his first attempts in 1936 to the realization of Continuity in Frankfurt, he produced dozens—if not hundreds—of variations on the infinity loop, exploring differences in scale, material, and environment.

But how are we to understand the topology of Continuity? Is it a Möbius strip, a variation on an endless loop, or simply an artistic, sculptural interpretation of an infinite motif? In short, how did he do it? Even though Bill left some explanations, our own experiments and consultations with mathematicians did not yield an immediate understanding of how the sculpture was conceived and modeled.

Bill wrote: “In Continuity, I intended to achieve something similar: a single surface, a single boundary. Theoretically, a double helix, twisted twice by half a turn and fused with one twisted only half a turn, should have resulted in something akin to an ‘infinite loop,’ possessing only one surface and one boundary. For reasons unknown to me, this did not happen, so Continuity, despite the theoretically expected outcome, consists of two parallel bounded surfaces composed of infinitely many straight lines. This explanation may sound somewhat dry and simplistic, yet it is the only thing I can say about it myself.”

The interesting thing to note is that Bill thought he had invented a completely new shape. Even more curious was that he discovered it by twisting a strip of paper, just as Möbius had done many years previously. "Not long afterwards, people began to congratulate me on my fresh and original interpretation of the Egyptian symbol for infinity and the Möbius ribbon. I had never heard of either of them. My mathematical knowledge had never gone beyond routine architectural calculations, and I had no great interest in mathematics."

Barry Cipra (mathematician, Minnesota, USA): “Bill’s intuition that a single half twist connected to a double half twist should result in a single-sided, single-edge surface is quite correct; depending on the handedness of the twists, the loop you wind up with has either three twists or one. So it’s a mystery how he managed to add 1 and 2 and get

4.” (link)

By carefully examining the sculpture itself and creating a 3D model using 360° video and photography (fig. 1), and by comparing it with Bill’s metal sculpture from 1945–46—used by him as a 1:20 prototype (fig. 2)—we were able to analyze the work and conduct a series of spatial experiments. One of these resulted in an origami model that follows the original topology step by step and, surprisingly, resolves into a clear infinity-loop construction (fig. 3). Building on this, we developed a responsive interactive model that clearly demonstrates the topology of the *Continuity*. The result is striking: a pure mathematical model transforms into a work of art.

+ Max Bill Quotes

“There is continuity in our purpose. However our purpose is also to see—a nd to see clearly. We [designers] are not working for the sake of machines, for some impersonal, superordinate entity such as ‘industry’ or ‘state’ or ‘community’. We are working for the wellbeing of every person, every individual, as part of a community. <...> This purpose, this new humanism, is what I’d like to see as continuity”. *Max Bill, “Continuity and change”, 1953*

“the mysterious dimension of mathematical problems, the inexplicable dimension of space, the distance or closeness of infinity; a space which begins on one side and ends in a changed manner on the other — yet simultaneously the same — side; limitation without a fixed boundary, the variety that nevertheless forms a single whole; and the uniformity that ts changed by the presence of a single accent of power; the vibrations and illumination of adjacent particles of colour; the force field comprising a host of variables; the parallel lines that cross each other and the infinity

that returns within itself as the present; and next to it the square again in all its strength; the straight line unruffled by any relativity, and the curve forming a straight line at each of its points — all these things which seemingly have nothing to do with everyday human needs are nevertheless of immense importance when they become art. these forces we are dealing with are elementary forces on which every human order is based and which are contained in every order we can perceive.”
Max Bill